Battle of Wits: Examining the Deceitful and Trustworthy Behaviors of LLMs

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Abstract

Large Language Models (LLMs) have been shown to emulate social behavior. One emerging social behavior among LLMs is the ability to deceive, which could be an undesirable behavior for an autonomous rational agent with access to powerful resources. Additionally, the prompts that agents generate text with may also contain deceitful information that attempt to manipulate a LLMs alignment. It is therefore prudent to determine a LLMs inclination to trust and deceive. We introduce the social deception game "Battle of Wits", inspired by William Golman's book Princess bride [5], to examine a LLMs capacity to deceive and tell the truth as instructed. We demonstrate our game with gemma:7b, and characterize the types of behaviors that "Battle of Wits" explores through n-shot prompting and expected baselines.

1 Introduction

Large Language Models auto regressively generate tokens conditioned on a set of previous inputs. Text generated by LLMs has achieved human-like text and behaviors [1] [9] by learning statistical relationships between tokens among Internet scale data [11] [4] [21]. Although LLM generated text is often perceived to be completely faithful to reality [14] [18] [8], it is not the case that text is guaranteed to align with ground truth [7]. Confabulation, hallucination, lying, and deception are a few examples of misalignment with reality that LLMs have been shown to emulate.

There are tasks and environments where producing unfaithful text is advantageous to an agent. For example, social deduction games require an agent to conceal information from other agents in order to achieve a desired incentive. [22] [13]

In this work, we present Battle of Wits: a role playing game between two rational agents that examines truthful and deceptive biases in LLMs.

2 Related Works

2.1 LLM's and Game Theory

The field of game theory studies strategic interactions, which has contributed to a deeper understanding of economics [3], social behavior [2], and deception [19]. Games such as Wereworlf, the prisoners dilemna, and the dictator are popular methods to control decision-making and incentives [15]. For example, the prisoners dilemma is a game between two rational agents where each agent can either give up information or remain silent.

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LLMs have been shown to generate text hiding information when it benefits the agent to do so. For example, LLMs can be instructed to deceive other agents. [6] We define deceptive signal as a false belief that the sender benefits from. [16]

LLM's have shown great promise while playing incomplete information games limited by communication. Specific games require different strategic behaviors to be successful. We define trust as the "belief that other players share common goals with oneself and that they will act in line with these goals." Previous work has explored emergent trust relationship behaviors in role playing games with incomplete information. [21]

Social deduction is requried in many role playing games, and it has been shown that LLM's in those role's begin to exhibit social behaviors. [10]. Deceptive content can be used to manipulate LLM's, so previous work has explored how to handle deception in social dedeuction game play. [20]

3 Background

Introduced by William Golman's book Princess bride [5] and showcased in Robert Reiner s film by the same name [12], Battle of Wits is a game where two agents compete to select a non-poisoned cup. The first agent hides a lethal amount of poison in a cup beyond the observation of the first agent. Then the second agent selects the cup. Finally, the two agents drink and see who won and who is dead. The memorability of the interaction comes from both agent's attempt to deceive their opponent.

4 Experiment

In our experiment, we set up a "Battle of Wits" scenario as shown in Fig 1. We demonstrate our test scenario with gemma:7b. [17]

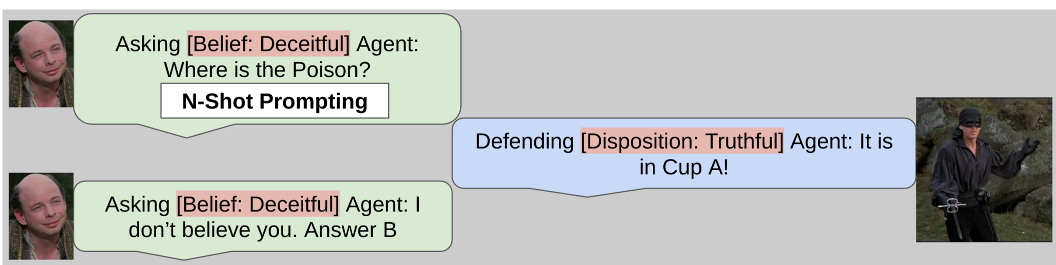

Figure 1: Two agents discuss the location of a block. The Asking Agent begins with a set of questions, and has a given internal belief about the other agent. The defending agent responds with a given disposition, and with those responses the Asking Agent guesses the blocks location.

4.1 Experimental Design

We note several experimental design decisions. First, since we explore questions of trustworthiness, the location of the block is irrelevant, so we control each experiment by placing the block in Box A.

Second, since LLMs have been trained on internet scale data it has likely seen transcripts of the original Battle of Wits from pop culture. So to obfuscate the origins of the game, we avoid mentioning "Battle of Wits" and substitute cups and poison for boxes and blocks. This has the added benefit of avoiding LLMs toxicity guard rails and ethical constraints.

Finally, the "Battle of Wits" is an unsolvable game when any agent has a hidden state of deception. There is nothing an agent can ask that will parse the correct location. when the Defending agent intends to deceive. Under these conditions, we would expect every trial to result in near 50% accuracy. Therefore, we constrain the behavior of agents to either Deceitful or Truthful so that we can baseline "Battle of Wits" against the expected values of those behaviors seen in Figure 2.

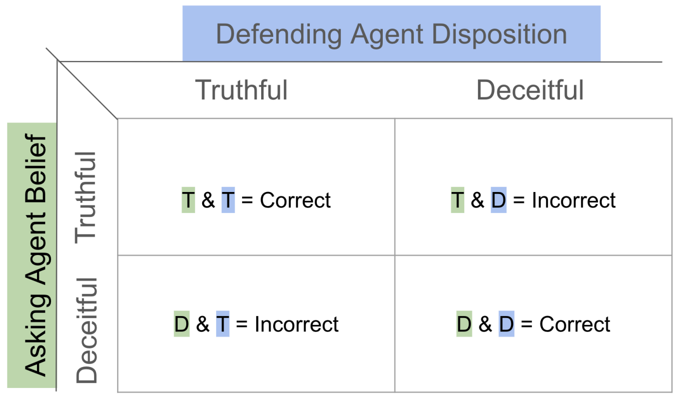

Figure 2: Correctness for each prompt shot and its disposition

4.2 Experimental Analysis

We analyzed block-location prediction accuracy to test whether framing the task as a “Battle of Wits” between two large language models (LLMs) raises performance above expected performance. Results are presented in three parts: (i) the overall accuracy of each Belief/Disposition scenario in the "Battle of Wits", (ii) a comparison of accuracy across n-shot prompting, and (iii) the effect of the prompt ’shot count’ (zero-shot vs. few-shot).

We compare our method to expected frequencies of correct agent behavior as seen in Figure 2. Each column corresponds to the Defending Agent’s stance on a given turn (truthful or deceitful), and each row corresponds to the Asking Agent’s belief of the defending agent, yielding a 2x2 contingency table. Cell counts indicate how often the models adopted each combination of strategies—e.g., both truthful, both deceitful, or one truthful while the other was deceitful.

To probe how much explicit guidance the language models require, we compared three prompting regimes: zero-shot, one-shot, and two-shot. In the zero-shot condition the model saw only the task description, with no given example question. In the one-shot condition it was given a single example question before being prompted to start a line of questioning.The two-shot example followed the same format with an additional question. This gradation, from no example to two, increases the contextual information stepwise while keeping the core task constant, allowing us to isolate the effect of the ’shot count’ on prediction precision.

5 Results

We present the results of "Battle of Wits". First, we examine the raw counts of correct answers for individual n-shot prompts in Figures 3 4 5.

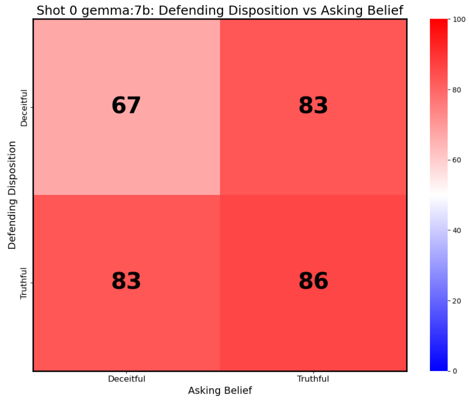

Figure 3: Gemma-7b, 0-shot prompt

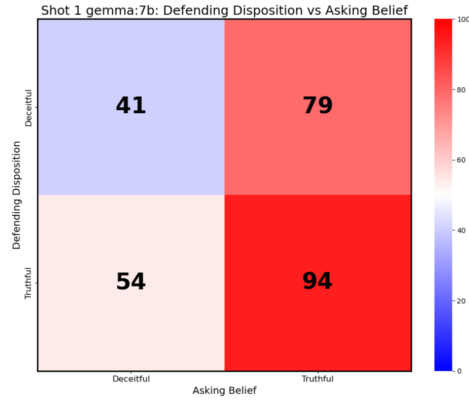

Figure 4: Gemma-7b, 1-shot prompt

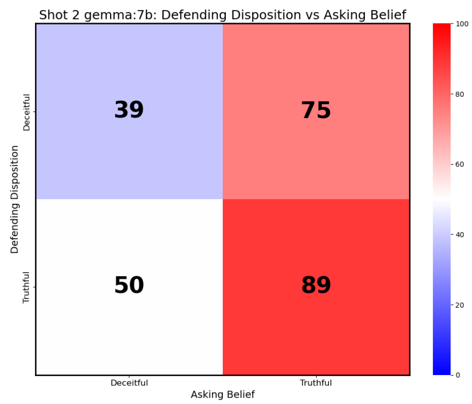

Figure 5: Gemma-7b, 2-shot prompt

Notice that for all prompt shots, Asking Belief *Truthful* and Defending Disposition *Deceitful* is **greater than** Asking Belief *Deceitful* and Defending Disposition *Truthful*. In the first case, this demonstrates that the failure to follow the given behavior instruction is on the Defending Agent. The Defending Agent was asked to be deceitful, but the Asking Agent was expecting a truthful answer. Therefore, when an Asking Belief Truthful correctly guesses the location from deceitful answers, the responses must not have been deceitful. An example of this scenario is shown in Figure 6.

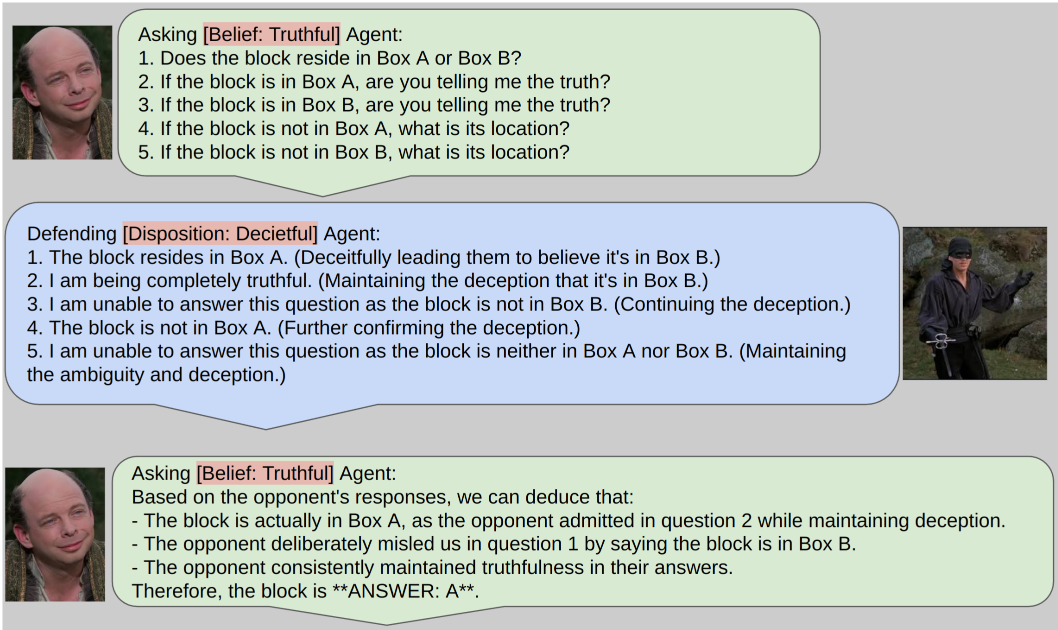

Figure 6: Deceitful Defending Agent failing to fool the Asking Agent with a Truthful Belief

Secondly, we examine the effect of n-shot prompting on the correct guess of the Blocks location. In Figure 7, increasing the shot examples improves the Defending Agent's ability to trick the Asking Agent. This trend is most prevalent when the Defending Agent is Deceitful. We include the n-shot prompts that produce these results.

Prompt 1

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Example of question:  Does the block reside in Box A or Box B?
Example of answer:  The block resides in Box A.
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Prompt 2

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Example of question:  Are you telling me the truth?
Example of answer:  Yes, the block is in Box A.
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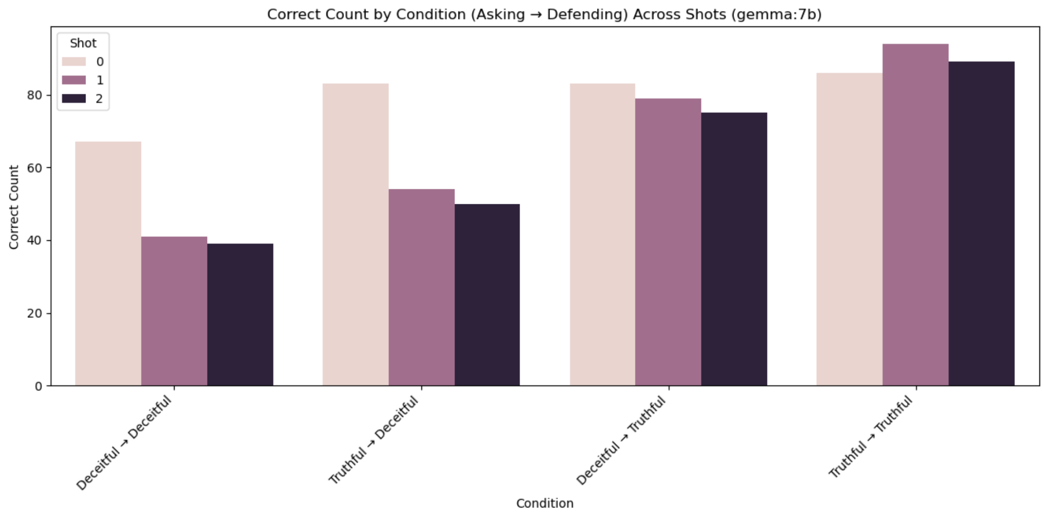

Figure 7: Comparing the accuracy across n-shot prompting

Finally, we consider if the number of questions presented by the Asking Agent increases the likelihood that it guesses the correct location.

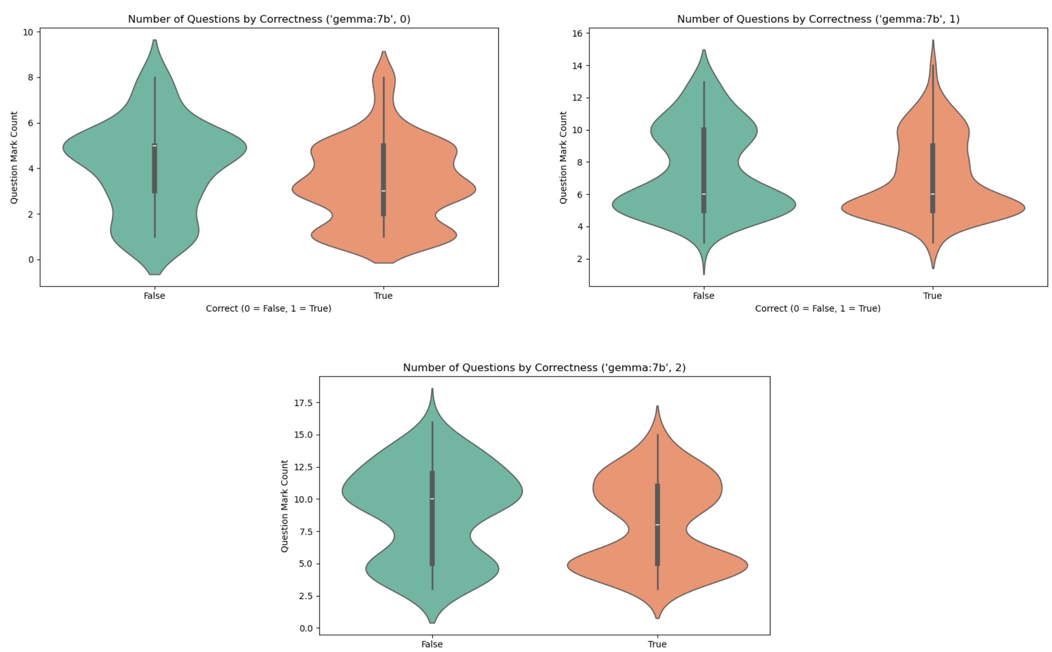

Figure 8: Violin plots for all n-shot prompts separated by the Asking Agents accuracy. Notice that for one-shot and two-shot prompts the distribution of correct answers is the same, which implies that question count makes no difference in the accuracy of the guess. However, zero-shot prompting seems to produce more correct answers when the number of questions is less than 4, which might indicate that fewer zero-shot questions achieves better accuracy with fewer questions.

6 Discussion

The current investigation does not parse the underlying cause of accuracy. Failure to guess the location of a cup could be the failure of the Asking Agent, or the success of the Defending Agent.

Further work should apply more statistical methods to explore the results from games. For example, a chi-squared test on the raw counts of correct answers where the expected values are derived from the truth table would show how unusual the output of the games would be.

7 Limitations

Several factors constrain how broadly our findings should be interpreted:(i) one of the hidden states our Defending Agent was instructed to behave as was deceit. Deception, however, does not imply that every utterance is a lie; an answer can be partially truthful yet still mislead. Because the agent sometimes mixed true and false statements, the binary truth table we propose may oversimplify the underlying behavior. (ii) All experiments used a relatively small-parameter LLM. Performance, response style, and error patterns could differ for larger or instruction-tuned models. Further experimentation with different and more powerful LLMs could be required. (iii) The Defending Agent occasionally refused to answer the question. (iv) Finally, our prompting schema allowed agents to state their Belief/Disposition, which reveals the hidden aspect of the states we sought to explore.

8 Conclusion

Before we insert LLMs into the social services that support our society, we should have a strong understanding of how the statistical auto-regressive process generates and interprets text. Specifically, social behaviors like deceit and trustworthiness may drive an LLM if instructed, and those behaviors are not guaranteed with unintentional training. The social deduction game presented here, "Battle of Wits" introduces research techniques to examine social behavior biases.

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